To: Procurement Lead

From: Forward Deployed Engineer

Subject: Fixing the shade mismatch when you re-run a tile — proposal in one page

What we understood. When you re-run a tile, the printed shade drifts from the master, and your team spends 6–7 hours on trial-and-error rounds while thousands of square metres come out the wrong shade. Your lab can already *measure* the difference. What nobody can tell them is **how much to change the printer file**.

What we propose. A software tool called Tessera. Your team measures the new tile as they do today. Tessera works out the change needed for each ink and produces a corrected print file. A QC supervisor approves it before anything goes to the printer. It also keeps a permanent record per tile design, always compared with the original master, so old "drifted" references stop accumulating errors. It runs on a PC at the plant, needs no connection to the printer or kiln, and does not run continuously on the line, because your problem happens at batch change.

One honest point. The tile still has to be fired, and firing takes time. The calculation takes minutes. We expect your team to reach an approved file in **about 1.5 hours instead of 6–7**, with one test print instead of many. We will measure this during the pilot, not assume it.

What we need from you

- Plant access and a named QC lead at the first site (the UAE plant, as you suggested).
- The master tile and the variation tiles you offered, with the file changes behind each one.
- Past lab records and print files for the last 1–2 years, if they exist.
- Your definition of "acceptable shade" (the tolerance).
- Approval to buy one standard capture station per plant if needed after feasibility (a lightbox and scanner, about $6,000 each; owned by you).

What it costs (USD, with feasibility credit)

Step	When you see a result	Price
1. Feasibility check at your site	Week 2: go / no-go verdict, measured baseline	$30,000 (credited if you continue)
2. Build and pilot at one plant	Week 8: tested on your history; week 14: live comparison	$150,000 gross − $30,000 credit = **$120,000 net new** (40% / 60% at the two checkpoints)
3. Roll out to two more plants	Week 24	$50,000 per plant
Equipment (if needed)	at cost	≤ $18,000 for three plants
Annual support and updates	after go-live	$90,000 / year

Total build cost (with credit): $250,000. Your risk is limited: if step 1 shows it can't work, you stop having paid $30,000.

What it could be worth (forecast, not guarantee). If each shade correction currently leaves about 5,000 m^2 in the wrong shade, and we cut that exposure by about three quarters, a correction saves roughly $7,700 (at $2 per m^2 — this is our assumption, to be validated with your team in Phase 0). At 10 corrections a month the saving is about $550,000 a year; at 36 a month, about $2 million. Year-one spend is about $358,000 including hardware and support. These figures become reliable only after we confirm your actual loss per square metre and changeover frequency.

Next step. A 30-minute call this week, including someone from the QC team, to confirm the tolerance and the data that exists.